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THERMODYNAMICS 1(MEC 122) LECTURE NOTES BY Mohammed Bashir A.

FUNDAMENTAL CONCEPTS AND LAWS OF THERMODYNAMICS

Introduction

Thermodynamics is a branch of mechanical engineering concerned with the relationship between heat, work, energy, and temperature of engineering systems and those properties of matter that relate to heat and work. In a broader form, it is a science in which energy storage, transformation, and transfer are studied. Energy is stored as internal energy (associated with temperature), kinetic energy (due to motion), potential energy (due to elevation), and chemical energy (due to chemical composition). It is transformed from one of these forms to another and transferred across a boundary as either heat or work.

Mathematical equations are developed in thermodynamics that relate energy transformations and transfers to material properties such as temperature, pressure, or enthalpy. Substances and their properties thus become an essential secondary subject. Much of our work will be based on experimental observations organized into mathematical statements or laws; thermodynamics' first and second laws are the most widely used.

When engineers study thermodynamics, their main goal is frequently to analyze or build a large-scale system, such as a nuclear power plant or an air conditioner. The activity of the constituent molecules in such a system can be averaged into quantifiable variables like pressure, temperature, and velocity [1].

Thermodynamics comprises two words: ***thermo*** (heat) and ***dynamics*** (power/transfer of heat energy). Therefore, **Thermodynamics** is concerned with the interactions of a system and its surroundings, or one system interacting with another. A system interacts with its surroundings by transferring energy across its boundary. Whenever there is an interaction between energy and matter, thermodynamics is involved.

Area of application: In terms of scope of application, thermodynamics is commonly encountered everywhere in life, such as in the pumping of blood in your body. **Household applications** include cooking stoves, pressure cookers, Air conditioning, iron, TV, Mobile, heater, refrigerators, etc. **Industrial applications** are automotive vehicles, airplanes, jet engines, nuclear power plants, Alternative Energy (solar, wind, Biomass), cooling, heating systems etc. Thermodynamics entails *four laws* known as Zeroth, First, Second, and Third laws of thermodynamics.

Thermodynamics, basically entails four laws or axioms known as Zeroth, First, Second and third law of thermodynamics.

1. The Zeroth law deals with thermal equilibrium and establishes a concept of temperature.

2. The First law throws light on concept of internal energy.
3. The Second law indicates the limit of converting heat into work and introduces the principle of increase of entropy.
4. The Third law defines the absolute zero of entropy.

These laws are based on experimental observations and have no mathematical proof. Like all physical laws, these laws are based on logical reasoning [2].

SYSTEM, BOUNDARY, SURROUNDINGS WORKING FLUID OF A SYSTEM

A thermodynamic **system** is defined as *a quantity of matter or a region contained within some closed boundary (surface).* The surface that separates the system from its surroundings is called the **boundary.** The mass or area outside the system is called the **surroundings (**Figure.1)**. The Boundary** of a system can be **real** one like that enclosing the gas in the cylinder of Figure.1 or an **imaginary** boundary like that of a nozzle opened from both ends (Figure 6). In the case of a nozzle, the inner surface of the nozzle forms the **real** part of the boundary, and the entrance and exit areas form the **imaginary** part since there are no physical surfaces there (Figure 6). Boundary may also be also a *fixed boundary (e.g., potato)* or *movable (e.g., piston cylinder).* Note that the boundary is the contact surface shared by the system and the surroundings. In Figure.1 the system is the compressed gas, the working fluid, and the system boundary is shown by the dotted line.

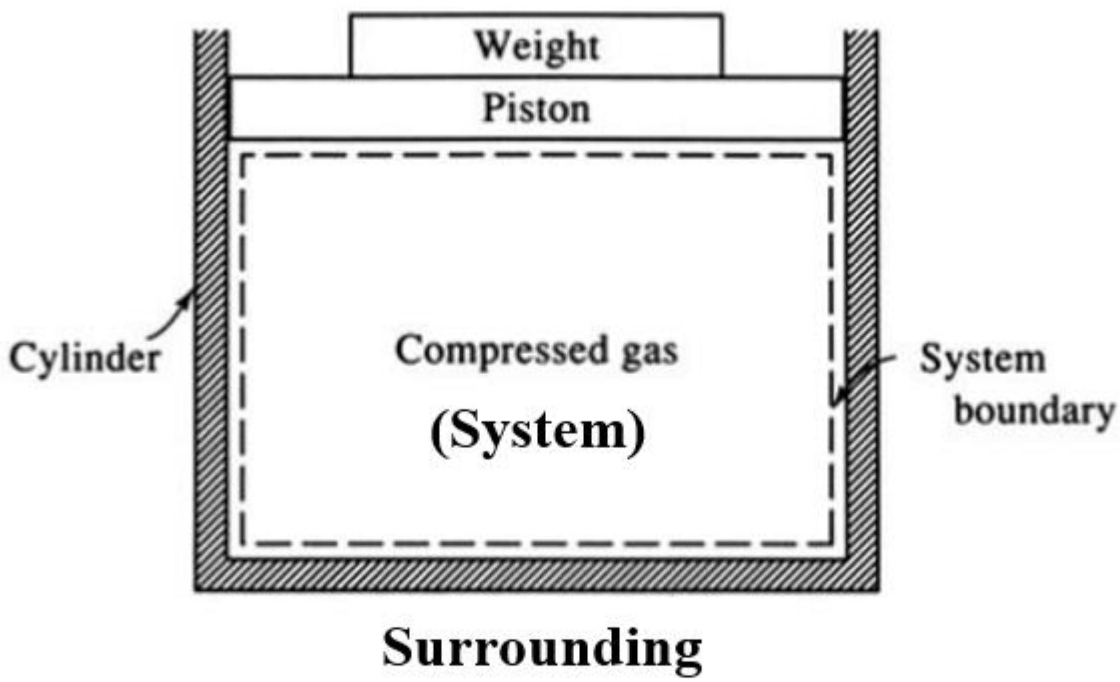


Figure.1 System, surroundings, boundary

A universe comprises a system, its surroundings, and the system boundary. We allow two essential interactions between the system and its surroundings:

- Heat can cross into the system (our potato can get hot), and
- Work can travel out of the system (our potato can expand).

The system boundaries can change; for example, the potato might expand on heating, but we can still distinguish the system and the surroundings.

The working fluid in a thermodynamic system is a liquid or gas that absorbs or transmits energy. Working fluid can be described as the **matter contained within the boundaries of a system**. In other words, it is a transport agent for energy and mass interactions. The working fluid can either be vapor or gaseous [3]. The working fluid used in the refrigeration cycle is called refrigerant. A working fluid can be a pressurized gas or liquid that actuates a machine. Examples include steam in a steam engine, air in a hot air engine, and hydraulic fluid in a hydraulic motor or cylinder. Heat engines and other cyclic devices usually involve a fluid to and from which heat is transferred while undergoing a cycle. The state of the working fluid is defined by specific characteristics known as properties [1].

TYPES OF THE THERMODYNAMIC SYSTEMS

After defining the system and boundary, observing the interaction between the system and its surrounding across the boundary is crucial. This interaction can be in form of mass and energy. Based on the interaction across the boundary, the system can be categorized into three types: **closed**, **isolated**, and **opened systems** (Figure.2), depending on whether a fixed mass or a fixed volume in space is chosen for study.

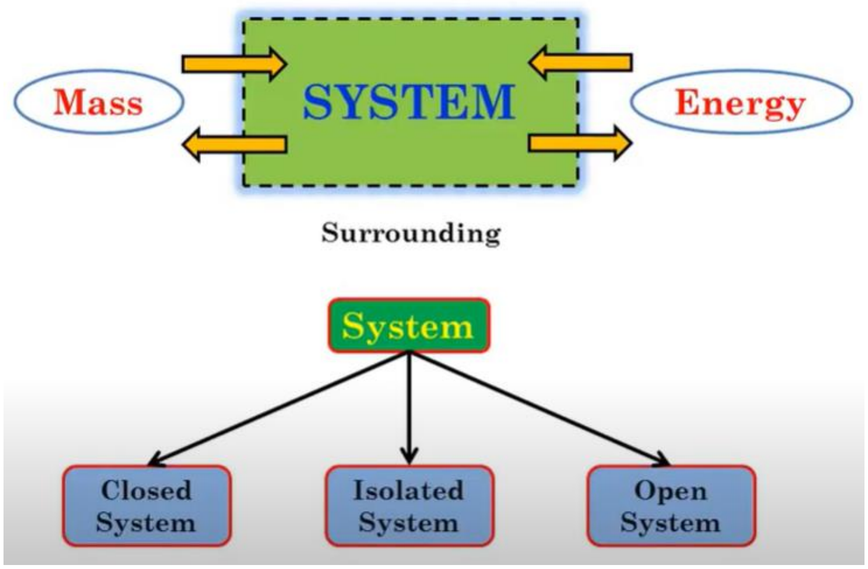


Figure.2 Types of the thermodynamic system in detail

i. **Closed system** (control mass)

A closed system is a type of system that has only energy interaction with the surroundings but no mass interaction. For example, consider a tightly closed cylinder device (Figure 3). In this system, no mass can transfer in or out of the boundary, but energy can move across the system boundary in the form of Heat or Work. It is also known as **control mass** because it consists of a fixed amount of mass, and no mass can cross its boundary. Note that the volume of a closed system does not have to be fixed.

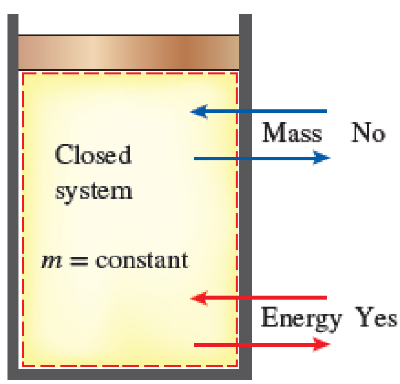


Figure 3 A closed system mass cannot cross the boundaries of a closed system, but energy can

An example of a closed system with a moving boundary considers a tight piston-cylinder device (without valves), as shown in Figure 4. Let's say we would like to find out what happens to the enclosed gas when it is heated. Gas inside the cylinder is our system. The inner surface of the piston cylinder forms the boundary. It is interesting to observe that this boundary is a moving boundary and the real boundary, and since no mass is crossing the boundary, this is called a closed system. Note that here energy is crossing the boundary in the form of heat, and the part of the boundary may move because of the expansion of the gas. Everything outside the gas, including the piston cylinder, is called the surrounding.

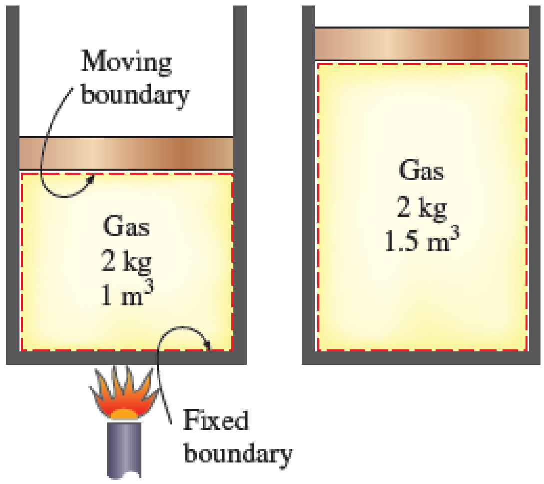


Figure.4 A closed system with a moving boundary

Keep some amount of fluid in the cylinder and close all the valves. By applying heat to the bottom of the cylinder, it will start to expand the fluid (Figure 4), and as a result, the temperature of the fluid will gradually increase, and that heat will be emitted outside. So, we added energy into the system and energy transferred too. Hence, there is energy transfer, although there was no mass transfer. Other examples include water in a closed pressure vessel, cooking in a pressure cooker, a Car Battery, a Piston cylinder, a can of cold drinks.

ii. Isolated system

Isolation means being totally separated from something. An isolated system is not influenced by its surroundings so that there will not be any mass or energy transfer between the system and surrounding (Figure.5). Thermos Flask is an example of an isolated system. We all have flasks in our homes.

a. What do we do with it? We stored hot water, hot tea, or hot milk for a long time.
b. How is it possible? Because this flask is well **insulated**, it helps prevent heat energy from going out, and we get hot water.

c. But have you observed the temperature of hot water after a long time is reduced and later on it will be same as normal temperature. Why? Because the insulation is not perfect, if the insulation is perfect, that means it will not allow any heat transfer across the boundary; then there will not be any mass transfer as well as energy transfer. Thus, **an isolated system is a special case of a closed system**

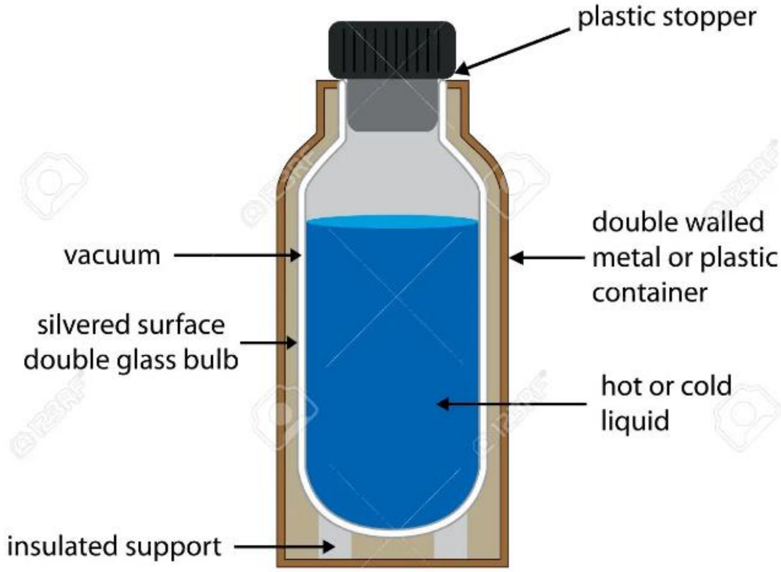
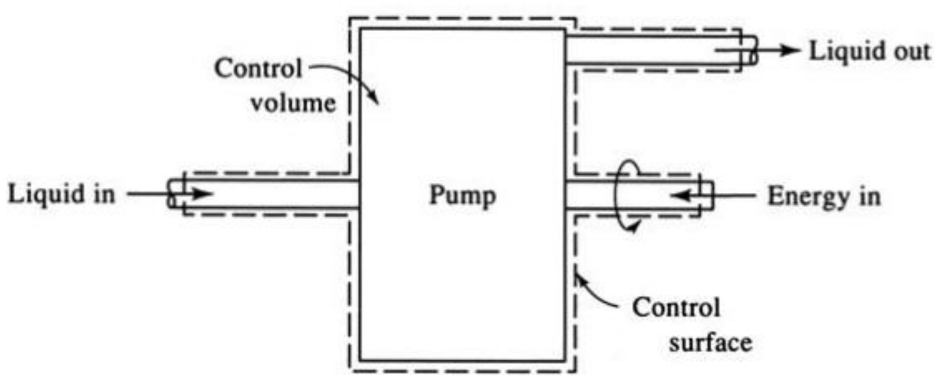


Figure.5 An Isolated system thermos Flasks

iii. Open system (control volume**)**

Refer to Figure 6. An open system is one in which matter flows into or out of the system. Most of the engineering systems are open, such as a pump, compressor, and Boiler, turbine, or nozzle, water in a pipe, a water heater, Engine oil in the engine, a car radiator Water Pump, the liquid in the heat exchanger. An open system, also known as control volume.

, A control volume can also involve heat and work interactions just as a closed system, in addition to mass interaction.



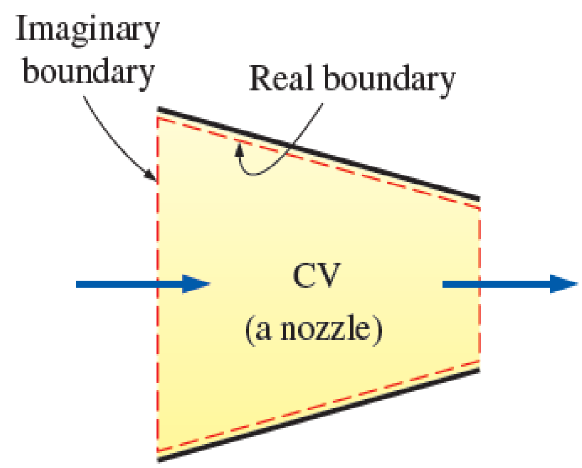


Figure 6 A control volume (CV) with real and imaginary boundaries

THERMODYNAMIC STATE, EQUILIBRIUM,

State' of a system indicates the **specific condition** of the system as described by its properties.. At a given *state*, all the properties of a system have fixed values. Thus, if the value of even one property changes, the state will change to different one.

Thermodynamic system undergoes changes due to the energy and mass interactions. Thermodynamic state of the system changes due to these interactions. The mode in which the **change of state** of a system takes place is termed as the **process** such as constant pressure process, constant volume process etc. The succession of states passed through during the change of state is called the **path of the change of state.** Thus, a process is a transformation from one state to another **(**Figure 7**),** e.g. constant pressure process**.** When all the properties of a system have definite values, the system is said to exist at **a definite state.**

In an **equilibrium** state, there are no unbalanced potentials (or driving forces) within the system. A system in equilibrium experiences no changes when it is isolated from its surroundings.

Types of equilibrium

i. **Thermal equilibrium**: when the temperature is the same throughout the entire system.
ii. **Mechanical equilibrium**: when there is no change in pressure at any point of the system. However, the pressure may vary within the system due to gravitational effects.
iii. **Phase equilibrium**: in a two phase system, when the mass of each phase reaches an equilibrium level.
iv. **Chemical equilibrium**: when the chemical composition of a system does not change with time, i.e., no chemical reactions occur.

PROCESS AND CYCLE

Any change a system undergoes from one equilibrium state to another is called a ***process,*** and the series of states through which a system passes during a process is called a *path*

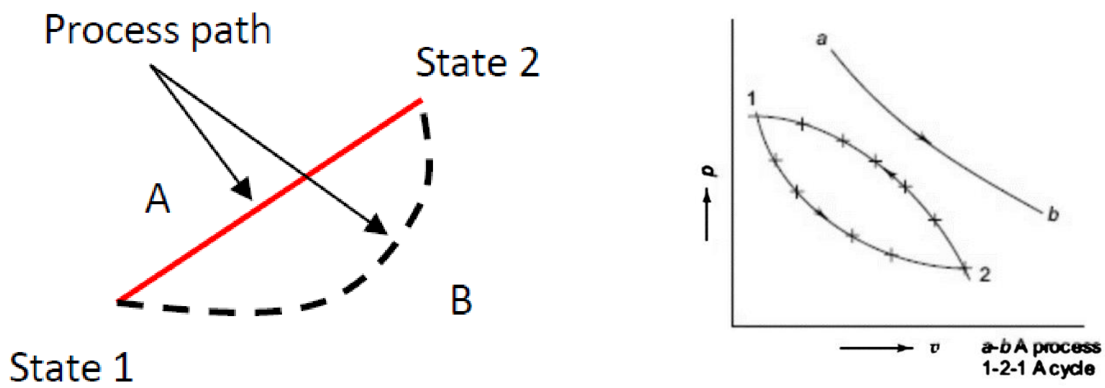


Figure 7 A state, process and a cycle

Note: To specify a process, initial and final states and path must be specified.

Any process or series of processes whose end states are identical is termed a **cycle,** Thus, a **thermodynamic cycle** is a series of processes (state changes) in which a working fluid begins at an initial state, moves through one or more intermediate states, and then returns to its initial state (Figure 7**)**. If network is produced by the cycle, then it is labeled a power cycle. If work is added to the cycle and heat is pumped from a cold space to a hot space, then the cycle is a refrigeration cycle.

A closed system is said to undergo a cyclic process, or *cycle*, when it passes through a series of states in such a way that its final state is equal in all respects to its initial state.

QUASI-EQUILIBRIUM PROCESS

Quasi-static (Quasi-equilibrium): Quasi means '*almost*'. A quasi-static process is also called a *reversible process.* It can be viewed as a sufficiently slow process that allows the system to adjust itself internally and remains infinitesimally close to an equilibrium state at all times. Quasi-equilibrium process is an idealized process and is not a true representation of the actual process. We model actual processes with quasi-equilibrium ones. Moreover, they serve as standards to which actual processes can be compared [2].

To consider how a gas (or liquid) might be expanded or compressed in a quasi-equilibrium fashion, refer to Figure.8, which shows a system consisting of a gas initially at an equilibrium state. As shown in the figure, the gas pressure is maintained uniform throughout by a number of small masses resting on the freely moving piston. Imagine that one of the masses is removed, allowing the piston to move upward as the gas expands slightly. During such an expansion the state of the gas would depart only slightly from equilibrium. The system would eventually come to a new equilibrium state, where the pressure and all other intensive properties would again be uniform in value. Moreover, were the mass replaced, the gas would be restored to its initial state, while again the departure from equilibrium would be slight. If several of the masses were removed one after another, the gas would pass through a sequence of equilibrium states without ever being far from equilibrium. In the limit as the increments of mass are made vanishingly small, the gas would undergo a quasi-equilibrium expansion process. A quasi-equilibrium compression can be visualized with similar considerations.

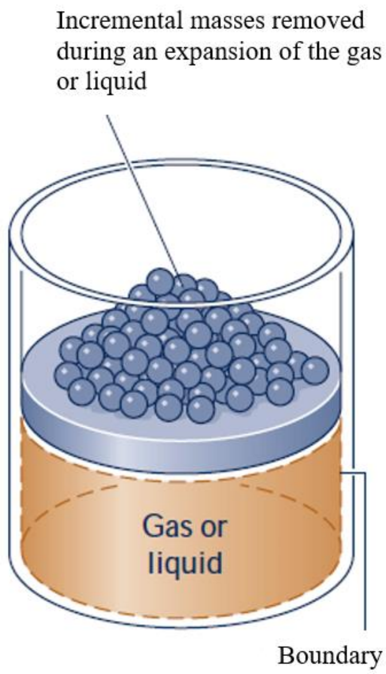


Figure.8 Illustration of a quasi-equilibrium expansion or compression

THERMODYNAMICS PROPERTIES

Any characteristic of a system which can be observed or measured is called a property of the system, e.g., volume, temperature, pressure etc. Thermodynamic properties are defined as characteristic features of a system capable of specifying the system's state. Some familiar properties are pressure P, temperature T, volume V, and mass m. The list can be extended to include less familiar ones such as viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, and even velocity and elevation. The thermodynamic properties of the system are further classified as the '**intensive** property' and '**extensive property'**.

Intensive properties

A quantity is called ***intensive*** when it is independent of the mass of a system or properties which have same value for any part of the system, such as temperature, pressure, velocity, density, Specific Volume, Specific Entropy, Thermal conductivity, Thermal Expansion, Compressibility and many more. (Figure 9). **They are not additive**

Extensive properties

Extensive properties are those properties which depend upon the mass of system and do not maintain the same value for any path of the system such as mass, volume, momentum, Energy (U), enthalpy (H), heat capacity, entropy (S), etc. **They are additive** [3],.

An easy way to determine whether a property is intensive or extensive is to divide the system into two equal parts with an imaginary partition, as shown in Figure 9. Each part will have the same value of intensive properties as the original system, but half the value of the extensive properties.

- Generally, uppercase letters are used to denote extensive properties (except mass *m*), and lowercase letters are used for intensive properties (except pressure *P* and temperature *T*).
- Extensive properties per unit mass, moles are called **specific properties e.g** specific volume ($v = V/m$) and specific Energy ($e = E/m$). In this case, we speak of a ***specific quantity*** or a ***reduced extensive quantity***.

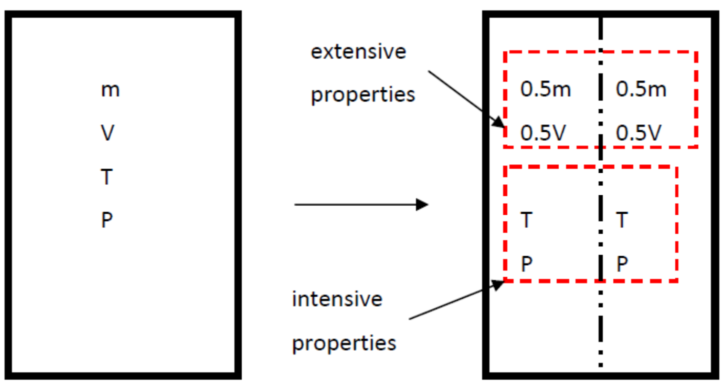


Figure 9 intensive and extensive properties of a system.

DEFINITION OF THE THERMODYNAMICS PROPERTIES

1. **Temperature**

The temperature (T), is a thermal state of a body which distinguishes a hot body from a cold body. Thus, temperature is a pointer for the direction of energy transfer as heat

Instruments for measuring ordinary temperatures are known as thermometers and those for measuring high temperatures are known as pyrometers.

It has been found that a gas will not occupy any volume at a certain temperature. This temperature is known as absolute zero temperature. The temperatures measured with absolute zero as basis are called **absolute temperatures**.

Absolute temperature is stated in Kelvin scale. The point of absolute temperature is found to occur at 273.15 K below the freezing point of water. Then, Absolute temperature =Thermometer reading in °C + 273.15. Another scale is the Celsius scale.

The concept of temperature was further refined by the introduction of the ***zeroth law*** of thermodynamics. It is called this because it was introduced about 100 years after the first and second laws.

2. Pressure

Pressure, P is defined as the normal force excreted per unit area of the surface within the system.

$$P = \frac{F}{A} \quad \frac{F}{A} = \text{Pa} \tag{1}$$

In fluids, gases and liquids, we speak of *pressure*; in solids this is *stress*. For a fluid at rest, the pressure at a given point is the same in all directions See Figure 10.

For engineering work, pressures are often measured with respect to atmospheric pressure rather than with respect to absolute vacuum.

$$P_{abs} = P_{atm} = P_{gauge} \tag{2}$$

In SI units the derived unit for pressure is the Pascal (Pa), where1Pa $= \frac{\text{N}}{\text{m}^2}$. This is very small for engineering purposes, so usually pressures are given in terms of kilopascals (1 kPa = 10^5 Pa), megaPascals (1 MPa = 10^6 Pa), or bars (1 bar = 10^5 Pa) and the standard atmosphere, where 1 atm = 101 325 Pa

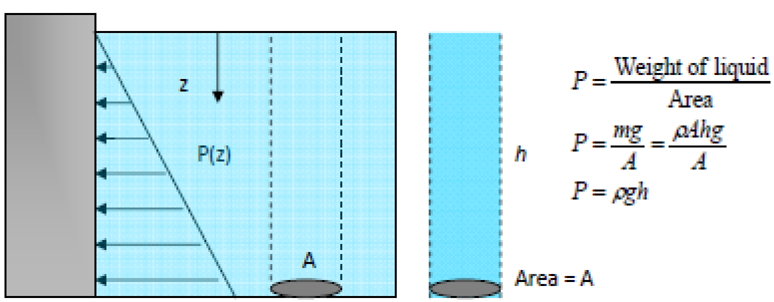


Figure 10 Pressure of a fluid at rest increases with depth (due to added weight), but constant in horizontal planes

3. Specific Volume (v) and Density (ρ)

The ***specific volume*** of a system is the volume occupied by the unit mass of the system. The symbol used is v and unit is, m^3/kg. The symbol V will be used for volume. (Note that specific volume is *reciprocal of density*). Mathematically as in equation 3b.

$$v = \frac{volume}{mass} \tag{3}$$

Units are $\mathbf{m^3/kg}$

It represents the inverse of the density, $v = \frac{1}{\rho}$ (3b)

4. **Internal Energy (u)**

Internal energy (U) - The property of a system covering all forms of Energy arising from the internal structure of the substance, SI Unit **kJ**. Specific energy (u), is $\frac{E}{\mathrm{m}}$. SI Unit **kJ/kg**

5. **Enthalpy**

Enthalpy (H) - A property of the system conveniently defined as H = U+ PV where U is the internal Energy. SI Unit **kJ.** Specific enthalpy, h=$\frac{H}{\mathrm{m}} = u + Pv$, SI Unit **kJ/kg**

6. **Entropy**

Entropy (S) - The microscopic disorder of the system. It is an extensive equilibrium property. SI Unit **kJ** but specific entropy (s) become intensive. SI Unit **kJ/kg.**[4]

5. **Specific Heats**

The ***specific heat*** is defined as the Energy required to raise the temperature of a unit mass of a substance by one degree. In general, this Energy depends on how the process is executed. In thermodynamics, we are interested in two kinds of specific heats**:** ***specific heat at constant volume*** $\boldsymbol{C_V}$and ***specific heat at constant pressure***$\boldsymbol{C_P}$.

Physically, the specific heat at constant volume $\boldsymbol{C_V}$ can be viewed as *the Energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant.* The Energy required to do the same as the pressure is maintained constant is the specific heat at constant pressure$\boldsymbol{C_P}$. Specific heat at constant pressure$\boldsymbol{C_P}$ is always greater than $\boldsymbol{C_V}$ because at constant pressure the system is allowed to expand, and the Energy for this expansion work must also be supplied to the system.

LAWS OF THERMODYNAMICS

i. Zeroth law of thermodynamics

If an object with a higher temperature comes in contact with a lower temperature object, it will transfer heat to the lower temperature object. The objects will approach the same temperature, and in the absence of loss to other objects, they will maintain a single constant temperature. Therefore, thermal equilibrium is attained.

The "**Zeroth law"** states that if two systems are at the same time in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

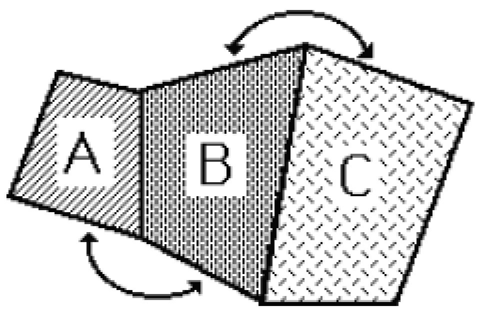


Figure11 Analogy of the Zeroth Law of Thermodynamics

If A and C are in thermal equilibrium with B, then A is in thermal equilibrium with B. Practically this means that all three are at the same temperature, and it forms the basis for comparison of temperatures.

In other words, the Zeroth Law states that:-*"two systems which are equal in temperature to a third system are equal in temperature to each other"*. This law deals with thermal equilibrium and establishes a concept of temperature.

ii. First law of thermodynamics

The first law of thermodynamics is the application of the conservation of energy principle. States that *Energy can be neither created nor destroyed during a process; it can only change forms.* This simply means that during an interaction, energy can change from one form to another but the total amount of Energy remains constant. This law throws light on concept of internal Energy.

Example of first law applicable to a closed system. Consider a closed system where there is no flow into or out of the system, and the fluid mass remains constant. For such system, the first law statement is known as the Non-Flow Energy Equation, or NFEE abbreviated, it can be summarised as follows:

$$\Delta U = Q - W \qquad 4$$

The first law makes use of the key concepts of internal energy (ΔU), heat (Q), and system work (W).

iii. Second law of thermodynamics

Second law of thermodynamics states that *heat will flow naturally from one energy reservoir to another at a lower temperature, but not in opposite direction without assistance*. This is very important because a heat engine operates between two energy reservoirs at different temperatures. Further the first law of thermodynamics establishes equivalence between the quantity of heat used and the mechanical work but does not specify the conditions under which conversion of heat into work is possible, neither the direction in which heat transfer can take place. This gap has been bridged by the second law of thermodynamics.

v. Third Law of Thermodynamics

The third law of thermodynamics states "When a system is at zero absolute temperature, the entropy of system is zero".

All the laws of thermodynamics are based on experimental observations and have no mathematical proof. Like all physical laws, these laws are based on logical reasoning.

QUSTIONS

Briefly answer the following questions:

1. What is thermodynamics? List 6 the application areas of thermodynamics.
2. Write short notes on the following: Thermodynamic properties with 5 examples with their SI units, and closed system, isolated system, and open system, extensive and intensive properties. Support your explanation with examples and sketches where necessary.
3. Briefly explain the term working fluid in thermodynamics with examples
4. Explain the following terms: (*i*) State, (ii) path, (*iii*) Process, (i*v*) Cycle and (v) specific heat at constant volume and pressure.
5. State the four laws of thermodynamics.
6. What is a quasi-static process?
7. State thermodynamic definition of work. Also differentiate between heat and work.
8. What is energy? What are different forms of it?

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